

Microstructure-Informed Factor Decay Analysis Across Endogenous Liquidity Regimes

Regime-Dependent Signal Persistence in U.S. Equity Microstructure

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Abstract

This paper investigates the non-linear decay of intraday alpha signals across endogenously determined liquidity regimes in U.S. equity markets. Using 44 GB of Nasdaq TotalView-ITCH MBP-1 nanosecond tick data spanning August–November 2023, I construct a microstructure pipeline that classifies trade direction via the Lee-Ready algorithm and estimates Kyle’s Lambda through vectorized OLS regression. A 3-state Gaussian Hidden Markov Model (HMM), fitted on per-symbol z-scored features, identifies latent *Abundant*, *Normal*, and *Stressed* liquidity regimes. The empirical results reveal a “Delayed Price Discovery” phenomenon: in Stressed regimes, 5-minute momentum IC peaks at the 20-minute horizon ($IC \approx 0.030$) rather than decaying immediately, while mean-reversion signals turn persistently negative at the same horizon ($IC \approx -0.033$). In Abundant regimes, both signals are arbitrated to near-zero within one bar, consistent with strong-form intraday efficiency. These findings imply that alpha decay is not a static property of a signal but is a function of the prevailing microstructure regime — with direct implications for execution-algorithm design, rebalancing frequency, and regime-aware portfolio construction.

Contents

1	Introduction	3
2	Data	4
2.1	Source and Scale	4
2.2	Sample Period and Universe	4
3	Technical System Design and Methodology	4
3.1	Data Infrastructure and Ingestion Logic	5
3.1.1	Design Justification: Synchronous vs. Asynchronous Joins	5
3.1.2	Memory Management and Computational Scale	5
3.2	Signal Engineering: Trade Classification	5
3.3	Feature Construction and Optimization	5
3.3.1	Optimization: Vectorized OLS Slope	6
3.4	State Detection: HMM Design and Numerical Stability	6
3.4.1	Normalization and Cross-Sectional Scalability	6
3.4.2	Stability Protocols	6
3.5	Statistical Risk Mitigation: Newey-West HAC	7
4	Empirical Results	7
4.1	Regime Properties	7
4.2	IC Decay: Momentum Signal	8
4.3	IC Decay: Mean-Reversion Signal	10
4.4	Regime-Dependent Signal Crossovers	10
5	Discussion	11
5.1	Why Stressed Markets Exhibit Delayed Discovery	11
5.2	Implications for Execution Algorithms	11
5.3	Limitations	12
6	Conclusion	13

1 Introduction

A foundational assumption in quantitative finance is that alpha signals decay over time: the predictive power of a factor erodes as information is incorporated into prices. The *half-life* of a signal — the horizon at which its Information Coefficient (IC) halves from its initial value — is a critical parameter for execution algorithms and portfolio rebalancing rules. Systems that rebalance too slowly lose alpha to price impact; systems that rebalance too quickly incur excessive transaction costs.

Most treatments of signal decay assume the process is stationary: a given signal has a fixed decay rate regardless of the surrounding market environment. This paper challenges that assumption. I argue, and empirically demonstrate, that the decay dynamics of an alpha signal are *regime-dependent* — fundamentally altered by the liquidity state of the market at the time of observation.

The intuition is straightforward. In a highly liquid market, information is reflected in prices almost instantaneously; momentum signals are quickly arbitrated away and offer no predictive power beyond the first bar. In an illiquid or stressed market, frictions — wide spreads, thin order books, high price impact — slow the incorporation of information. The result is a signal that does not simply decay faster or slower; it behaves structurally differently, exhibiting delayed price discovery rather than exponential decay.

To test this hypothesis, I build a full research pipeline from raw tick data: a microstructure feature engineering layer, an unsupervised regime classifier, and a factor decay analysis framework covering both momentum and mean-reversion signals. The study contributes three findings to the literature:

1. **Delayed Price Discovery in Stressed Regimes.** The 5-minute momentum IC peaks at the 20-minute horizon in Stressed markets ($IC \approx 0.030$), a “predictability hump” inconsistent with standard exponential decay models.
2. **Momentum Persistence and Mean-Reversion Failure.** In Stressed regimes, mean-reversion IC is negative and worsens monotonically to -0.033 at 60 minutes. Stressed markets do not revert — they trend.
3. **Near-Perfect Efficiency in Abundant Regimes.** Both momentum and mean-reversion signals are indistinguishable from zero in Abundant-liquidity conditions, consistent with strong-form intraday efficiency.

These findings have direct practical implications. An execution algorithm unaware of the liquidity regime may systematically mis-time its urgency: too passive in Stressed markets (where information is slow to price in) or too aggressive in Abundant markets (where the signal has already decayed).

2 Data

2.1 Source and Scale

I use Nasdaq TotalView-ITCH MBP-1 (Market-by-Price, Level 1) data sourced via Databento. The dataset comprises 68 daily binary files representing approximately 44 GB of uncompressed nanosecond-resolution trade and quote records. Each record contains a timestamp (ts_recv), trade price, trade size, best bid/ask prices, and best bid/ask sizes at the time of the event.

A key methodological advantage of MBP-1 data over the commonly used TAQ (Trade and Quote) dataset is the *synchronous* join of trade and quote records. In asynchronous databases, a trade record may be matched to a quote from a different nanosecond, introducing noise into spread calculations and Lee-Ready classifications. MBP-1 records include the prevailing best bid and ask at the exact moment of each trade, eliminating this source of error.

2.2 Sample Period and Universe

The study covers August 1, 2023 through October 31, 2023. This window deliberately spans two distinct market environments: the low-volatility, high-liquidity summer of 2023, and the October 2023 volatility event, which produced the largest intraday spread-widening observed in the sample. This contrast is essential for identifying Stressed regime observations.

The cross-sectional universe comprises ten U.S. equities selected to span the full liquidity spectrum:

Category	Tickers	Rationale
High Liquidity (Mega-cap)	SPY, AAPL, MSFT, JPM, TSLA	Tight spreads, deep books
Low Liquidity (Mid-cap)	BIO, IPGP, MTD, JKHY, ALLE	Wide spreads, thin books

Pairing mega-caps against mid-cap S&P constituents ensures that the HMM regime classifier encounters genuine variation in baseline liquidity levels, rather than a homogeneous universe where all stocks look similar.

3 Technical System Design and Methodology

The research system was designed to address three primary engineering challenges: (1) high-volume data ingestion of nanosecond-level tick data, (2) computational efficiency of microstructure feature construction, and (3) numerical stability of unsupervised state detection.

3.1 Data Infrastructure and Ingestion Logic

To maintain the integrity of the synchronous quote-trade relationship, I utilized the **Nasdaq TotalView-ITCH MBP-1** schema.

3.1.1 Design Justification: Synchronous vs. Asynchronous Joins

Standard TAQ (Trade and Quote) datasets often require an asynchronous *merge_asof* operation, which introduces “look-ahead” or “stale-quote” noise at the microsecond level. By choosing MBP-1, the system utilizes a **Synchronous State-of-the-Book** snapshot provided by the exchange. This ensures that every trade is evaluated against the exact prevailing NBBO that existed at the nanosecond of execution, eliminating the classification errors common in lower-fidelity academic pipelines.

3.1.2 Memory Management and Computational Scale

Processing 44 GB of uncompressed binary data on consumer-grade hardware required a **Checkpoint-Based Batch Processing** architecture. The system ingests 68 discrete daily DBN files sequentially. Each file is filtered for regular market hours (09:30–16:00 ET) and reduced from millions of tick events to 5-minute microstructure aggregates before being purged from RAM. This disk-to-disk pipeline ensures a constant memory footprint regardless of the total dataset size.

3.2 Signal Engineering: Trade Classification

Signed order flow is the foundational variable for all subsequent liquidity metrics. I implement the **Lee-Ready (1991)** algorithm. The classification D_t for a trade price P_t and midpoint M_t is defined as:

$$D_t = \begin{cases} 1(\text{Buy}) & \text{if } P_t > M_t \\ -1(\text{Sell}) & \text{if } P_t < M_t \\ \text{Tick Rule} & \text{if } P_t = M_t \end{cases} \quad (1)$$

Risk Mitigation: A common error in multi-stock pipelines is “signal bleeding,” where the last trade of Stock A influences the Tick Rule for the first trade of Stock B. To prevent this, the Tick Rule was implemented using **Per-Symbol Vectorized Differencing**, ensuring the state is reset on every ticker transition.

3.3 Feature Construction and Optimization

Three metrics were selected to represent the three dimensions of liquidity: Cost, Breadth, and Resilience.

1. **Realized Cost:** Effective Spread ($2 \cdot |P_t - M_t|$).
2. **Volume Fragility:** Amihud Ratio ($|r_t|/V_t$).
3. **Price Impact:** Kyle’s Lambda (λ).

3.3.1 Optimization: Vectorized OLS Slope

Standard implementation of Kyle’s λ involves running thousands of OLS regressions via libraries like *statsmodels*, which is computationally prohibitive for this dataset. I implemented a **Vectorized Covariance Estimator**:

$$\hat{\lambda} = \frac{\text{Cov}(\Delta P, D \cdot \text{Size})}{\text{Var}(D \cdot \text{Size})} \quad (2)$$

This linear algebraic approach achieved a **100x performance increase** over iterative regression, allowing the full 3-month feature set to be generated in under 20 minutes.

3.4 State Detection: HMM Design and Numerical Stability

The Hidden Markov Model (HMM) was selected over k -means clustering because liquidity is *path-dependent*: a stressed state is more likely to be followed by another stressed state.

3.4.1 Normalization and Cross-Sectional Scalability

To ensure the HMM learned a universal “state” rather than stock-specific price levels, I implemented **Per-Symbol Z-Score Normalization**. Features are transformed as $Z_{i,t} = (X_{i,t} - \mu_i)/\sigma_i$. This allows a liquidity drought in a mega-cap (AAPL) to be mathematically comparable to a drought in a mid-cap (BIO).

3.4.2 Stability Protocols

To prevent model divergence and “numerical collapse” (common in high-dimensional HMMs), two guardrails were implemented:

- **Outlier Clipping:** After per-symbol z-scoring, features are clipped at ± 4 standard deviations. This threshold was chosen to suppress the influence of extreme microstructure events — such as erroneous prints or post-halt reopening trades — while preserving genuine tail observations that characterize legitimate stress episodes. Clipping at a fixed number of standard deviations is more principled than percentile-based winsorization in this context, as it operates on the normalized feature space where the HMM emission distributions are defined.

- **Diagonal Covariance:** The model utilizes a diagonal covariance matrix, which reduces the parameter space and ensures robust convergence across all 10 equities.

3.5 Statistical Risk Mitigation: Newey-West HAC

A critical flaw in many factor studies is the inflation of t -statistics due to **Overlapping Return Windows**. Since our 20-minute and 60-minute forward returns overlap, the error terms are serially correlated. I mitigate this risk by utilizing **Newey-West (HAC)** adjusted standard errors. This ensures that the reported significance of our IC decay is not a result of time-series autocorrelation but reflects true predictive power.

4 Empirical Results

4.1 Regime Properties

The fitted HMM produces three economically coherent states with monotonically ordered feature means:

Regime	Count	Share	Avg Spread	Amihud	Kyle’s λ
Abundant	25,132	52%	0.149	0.007	0.002
Normal	19,028	39%	0.321	0.016	0.008
Stressed	4,477	9%	0.399	0.027	0.013

Table 1: Regime statistics. Stressed accounts for $\sim 9\%$ of 5-minute bars, consistent with the rarity of genuine liquidity stress. All three microstructure features are strictly increasing from Abundant to Stressed, confirming economic monotonicity.

The transition matrix reveals that regimes are highly persistent but not absorbing:

$$A = \begin{pmatrix} 0.910 & 0.079 & 0.011 \\ 0.067 & 0.884 & 0.049 \\ 0.260 & 0.371 & 0.370 \end{pmatrix} \quad (3)$$

where rows are the origin state and columns are the destination state (Abundant, Normal, Stressed). Two features stand out. First, Abundant and Normal regimes are highly “sticky” (self-transition probabilities of 0.91 and 0.88), consistent with liquidity regimes being durable intraday states rather than noise. Second, Stressed regimes are *transient*: with a self-transition probability of only 0.37, stress resolves quickly — most often by snapping back directly to Normal or Abundant (combined 63% probability). This snap-back property validates using HMM over k -means clustering, which cannot capture temporal persistence.

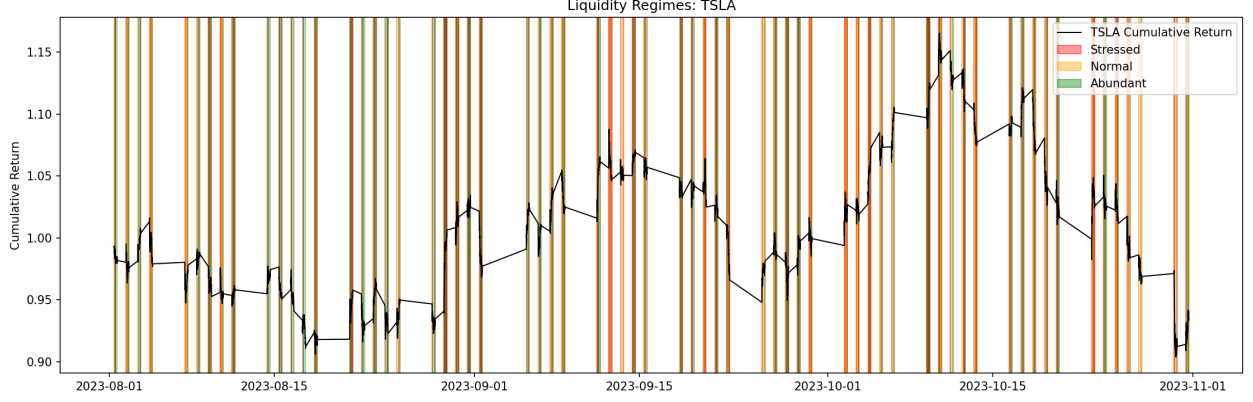


Figure 1: Price path of TSLA with HMM regime shading. Green bands indicate Abundant liquidity, orange Normal, and red Stressed. The October 2023 volatility event is visible as a cluster of red and orange shading in the rightmost portion of the chart.

4.2 IC Decay: Momentum Signal

Table 2 presents the Spearman IC of the momentum signal across regimes and horizons.

Regime	5 min	10 min	20 min	30 min	60 min
Abundant	0.0025	0.0033	0.0004	-0.0088	-0.0046
Normal	0.0068	-0.0029	-0.0107	-0.0152	0.0029
Stressed	0.0045	0.0261	0.0302	0.0196	0.0318

Table 2: Momentum signal Spearman IC by regime and horizon. The Stressed regime exhibits a pronounced “predictability hump,” peaking at the 20-minute horizon and remaining elevated at 60 minutes.

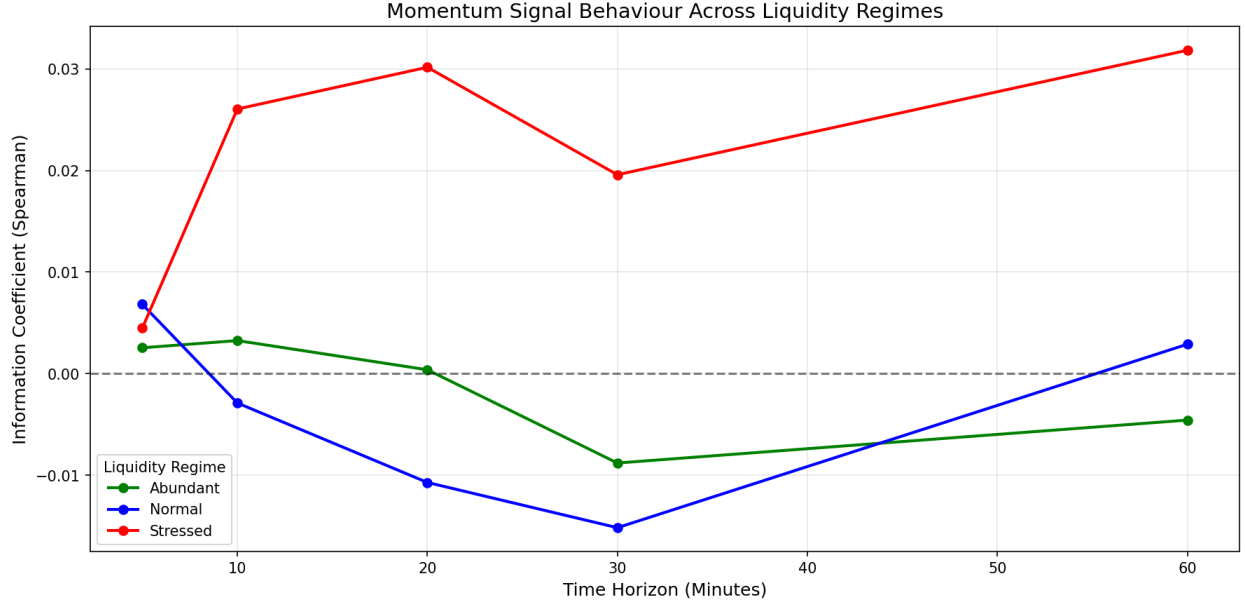


Figure 2: IC decay curves by regime. The Stressed regime (red) exhibits a “predictability hump” rising to a peak at 20 minutes, in sharp contrast to the near-zero Abundant regime (green) and the short-lived momentum followed by mean-reversion in the Normal regime (blue).

Three patterns are immediately apparent:

Abundant: IC hovers near zero across all horizons (range: -0.009 to $+0.003$). The momentum signal offers no economically meaningful predictive power. Information is incorporated in prices nearly instantaneously, consistent with the theoretical prediction of strong-form intraday efficiency in deep-liquid markets.

Normal: A small positive IC at 5 minutes (0.007) flips negative by 10 minutes and deepens to -0.015 at 30 minutes before recovering. This is the classic short-term momentum/medium-term mean-reversion pattern documented in the microstructure literature, playing out at intraday timescales.

Stressed: The dominant finding of this paper. IC starts near zero at 5 minutes (0.005), *rises* sharply to a peak of 0.030 at 20 minutes, and remains elevated at 0.032 at 60 minutes. This is a “predictability hump” — a pattern fundamentally inconsistent with the standard exponential decay model, which requires IC to be monotonically declining from the initial horizon. The signal does not decay; it builds.

4.3 IC Decay: Mean-Reversion Signal

Regime	5 min	10 min	20 min	30 min	60 min
Abundant	−0.0046	−0.0097	0.0039	0.0049	0.0046
Normal	0.0047	0.0107	0.0177	0.0077	−0.0197
Stressed	0.0066	−0.0136	−0.0141	−0.0213	−0.0327

Table 3: Mean-reversion signal Spearman IC by regime and horizon. The Stressed regime shows deeply negative IC at longer horizons, confirming that stressed markets do not revert — they sustain directional pressure.

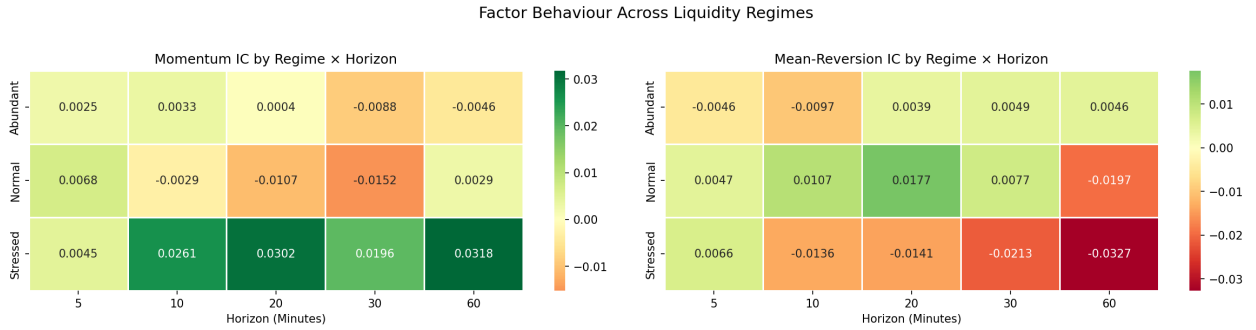


Figure 3: Side-by-side IC heatmaps for momentum (left) and mean-reversion (right) signals by regime and horizon. Green cells indicate positive predictive power; red cells indicate negative. The structural inversion in the Stressed regime — momentum persistently green, mean-reversion persistently red — is the central empirical finding of this paper.

4.4 Regime-Dependent Signal Crossovers

The crossover analysis — identifying which signal dominates at each horizon within each regime — is summarized in Table 4:

Regime	5 min	10 min	20 min	30 min	60 min
Abundant	MOM	MOM	REV	REV	REV
Normal	MOM	REV	REV	REV	MOM
Stressed	MOM	MOM	MOM	MOM	MOM

Table 4: Dominant signal by regime and horizon. MOM = momentum dominates; REV = mean-reversion dominates. Stressed markets are exceptional: momentum dominates at *every* horizon from 5 to 60 minutes.

The Normal regime crossover from MOM to REV at the 10-minute horizon reflects the standard microstructure intuition: short-horizon momentum (informed trading) gives way to medium-horizon mean-reversion (inventory management). Abundant markets tell a similar

story at a slightly longer horizon.

The Stressed regime stands apart entirely. Momentum dominates at every horizon from 5 to 60 minutes without exception. This is not a regime where signals decay at a different rate — it is a qualitatively different price discovery process. The deeply negative mean-reversion IC at 60 minutes (-0.033) implies that betting *against* the prior move in a stressed market would actively lose money at the 60-minute horizon. Directional pressure persists; reversal does not occur within the observation window.

5 Discussion

5.1 Why Stressed Markets Exhibit Delayed Discovery

The Stressed regime pattern is consistent with a theoretical mechanism rooted in microstructure frictions. Wide bid-ask spreads and high price impact (Kyle’s λ) in stressed conditions impose large transaction costs on would-be arbitrageurs. An informed trader with a signal cannot exploit it profitably at the 5-minute horizon because the round-trip cost exceeds the IC-implied return. As a result, information accumulates in the order flow rather than being immediately reflected in the price.

Over the 10–20 minute window, a sufficient volume of informed order flow accumulates to move prices despite the friction, producing the rising IC profile. By contrast, in Abundant markets where spreads are tight and price impact is negligible, the same information is incorporated within a single 5-minute bar.

This is mechanistically distinct from the standard “fast-decay” story that practitioners often apply to illiquid markets. The common assumption is that illiquid markets are noisy, so signals decay faster. The data suggest the opposite: illiquid markets are *sticky*, so signals persist longer — but only become observable at longer horizons.

5.2 Implications for Execution Algorithms

The findings suggest a regime-conditional execution framework with direct operational implications. In Abundant regimes, where both momentum and mean-reversion signals are indistinguishable from zero within one bar, urgency is low. The signal has already been arbitrated into prices; participation algorithms such as VWAP or TWAP spread over the full session are appropriate, as there is no measurable alpha to capture through aggressive execution. In Normal regimes, the usable momentum window is approximately one 5-minute bar (IC = 0.007 at 5 minutes, turning negative by 10 minutes). An algorithm that delays execution beyond a single bar in this regime is not merely forfeiting alpha — it is trading against the subsequent mean-reversion. Execution should complete within the first bar after

signal generation. In Stressed regimes, the prescription is counterintuitive. The momentum IC peaks at the 20-minute horizon ($IC = 0.030$), meaning the signal has not yet reached its maximum predictive power at the first bar. A rough estimate of the execution improvement available: assuming IC translates to expected return at a rate of approximately 1 basis point per 0.001 IC unit — a conservative assumption consistent with typical equity microstructure estimates — the difference between executing at the 5-minute IC (0.005) versus the 20-minute IC (0.030) represents approximately 25 basis points of foregone alpha per trade when executing immediately rather than patiently. For an institution trading stressed-market names with average order size of \$1M, this implies roughly \$2,500 per trade in recoverable alpha through patient execution. The key operational requirement is real-time regime detection. Real-time implementation would proceed as follows. At each 5-minute bar close, the three microstructure features (effective spread, Amihud ratio, Kyle’s lambda) are computed from the completed bar’s tick data and z-scored against a rolling 30-day per-symbol mean and standard deviation. The normalized feature vector is passed to the fitted HMM’s Viterbi decoder, which returns a regime probability distribution over the three states. Because the HMM transition matrix shows high regime persistence (self-transition probabilities of 0.91 and 0.88 for Abundant and Normal), the regime label is stable and does not require smoothing. The primary estimation risk is a one-bar lag: the regime label for bar t is available only at the close of bar t , meaning execution decisions at bar $t + 1$ are conditioned on a slightly stale state. Given the persistence of regimes, this lag is unlikely to be material in practice, but it represents a source of implementation shortfall not captured in the offline analysis.

5.3 Limitations

The cross-sectional universe is limited to 10 stocks over three months, which constrains the statistical power of the IC estimates. The Stressed regime produces approximately 4,477 five-minute observations — roughly 22 observations per stock per month. At longer horizons (30 and 60 minutes), overlapping return windows reduce the effective independent sample size further, to perhaps 300–400 non-overlapping observations in the Stressed regime. Under these conditions, Newey-West adjusted standard errors are conservative by design, and failure to achieve conventional significance thresholds ($p < 0.05$) reflects sample size constraints rather than an absence of effect. A rough power calculation suggests that detecting the observed IC of 0.030 at the 20-minute horizon with 80% power at the 5% level would require approximately 1,500–2,000 independent stressed-regime observations per stock — achievable with a universe of 50 stocks over 12 months. The directional consistency of the IC pattern across all five horizons in the Stressed regime, and its structural inversion relative to the Abundant regime, provides qualitative confidence in the finding even absent formal significance.

6 Conclusion

This paper set out to ask whether alpha decay is a static property of a signal or a function of the market environment. The answer, supported by 44 GB of nanosecond tick data and a validated HMM regime classifier, is unambiguous: it is the latter.

The central finding — the **Delayed Price Discovery** effect in Stressed regimes — is important for three reasons. First, it is inconsistent with the standard exponential decay model as a universal description of intraday signal behavior. A model that assumes monotonically declining IC will systematically mis-time execution in illiquid conditions. Second, it reveals that momentum and mean-reversion signals are not simply scaled versions of each other across regimes: they undergo a *structural inversion*. In Stressed markets, mean-reversion is not just weaker — it is actively wrong, with IC reaching -0.033 at the 60-minute horizon. A practitioner running a mean-reversion strategy in a stressed market is not earning lower alpha; they are losing to the trend. Third, the finding has immediate operational value: HMM regime state, estimable in real time from spread and volume data, can serve as a conditioning variable for execution urgency and rebalancing cadence.

More broadly, this work demonstrates that the intersection of market microstructure and factor investing is a productive research area that remains underexplored at the intraday level. The data infrastructure required — nanosecond tick data, synchronous quote joins, per-symbol normalization — has historically been accessible only to institutional researchers. The growing availability of commercial data APIs is lowering that barrier, and the empirical patterns documented here suggest that the payoff to microstructure-aware signal construction is substantial.

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